BRA-EUR - Towards Refining the Network Analysis

```
source(here::here("_chapter-setup.R")) # all standard packages and defaults
```

Overview

The idea is to develop a refined network analysis.
For this we load the data for 2024 for Brazil (i.e. ANAC data set) and Europe (NM flight table).

```
# -------------------- load the Brazilian network data -----------------------
cols <- c(
  "sg_empresa_icao",
  "nr_voo",
  "nm_mes_partida_real",
  "nr_mes_partida_real",
  "sg_icao_origem",
  "sg_icao_destino",
  "lt_combustivel",
  "sg_equipamento_icao",
  "nm_regiao_origem",
  "nm_pais_origem",
  "nm_regiao_destino",
  "nm_pais_destino",
  "nr_horas_voadas",
  NULL
)

bra_mvts <- read_csv("./data_proc/flights_ANAC_2024.csv"
                   #   , col_select = dplyr::any_of(cols)
```

```
                      , show_col_types = FALSE
                      ) |>
  rename(
    ADEP = sg_icao_origem,
    ADES = sg_icao_destino,
    OPR  = sg_empresa_icao,            # aircraft operator (ICAO)
    FUEL_BURN = lt_combustivel,
    TYPE = sg_equipamento_icao,        # aircraft type (ICAO)
    REGIAO_DEP = nm_regiao_origem,
    DEP_CNTRY = nm_pais_origem,        # departing country
    REGIAO_ARR = nm_regiao_destino,
    DES_CNTRY = nm_pais_destino,       # destination country
    FLT_DUR = nr_horas_voadas         # flight duration
)
```

```
New names:
* `` -> `...1`
```

```
# ---------------- Europe ---------------------------------------------------
# data is stored on local network drive
source(here::here("R","rqutils-zip.R"))
pth_to_data <- here::here(here::here() |> dirname(), "__DATA","NM-flight-table")
payload_zip <- "NM-flt-2024.zip"

check_zip_content(pth_to_data, payload_zip)  # check what we have in zip
```

```
                       Name    Length                Date
1 NM-FLT-2024-Q1.gz.parquet 166156360 2024-04-17 13:27:00
2 NM-FLT-2024-Q2.gz.parquet 226135599 2024-07-17 11:39:00
3 NM-FLT-2024-Q3.gz.parquet 228081776 2024-10-03 12:54:00
4 NM-FLT-2024-Q4.gz.parquet 197557076 2025-03-31 09:48:00
```

```
eur_mvts  <- read_zip(pth_to_data, payload_zip)
apts_meta <- arrow::read_parquet("./data/airport-icao-meta.parquet")

#| Utility function to prepare data -------------------------------------------
#|
prep_eur_network_data <- function(.eur_mvts, .apts_meta = apts_meta){
  this_mvts <- .eur_mvts |>
    dplyr::select(ADEP, ADES, FLTID = AIRCRAFT_ID, OPR = AIRCRAFT_OPERATOR
                 , REG = REGISTRATION, TYPE = AIRCRAFT_TYPE_ICAO_ID
```

```r
                  , FLT_DUR = FLT_DUR_3
                  , DATE = LOBT
                  , SAM_ID = ID
                  ) |>
    dplyr::mutate(DATE = date(DATE))

  # augment airport details
  this_mvts <- this_mvts |>
    dplyr::left_join(.apts_meta |>
                       dplyr::select(ADEP = ICAO, DEP_CNTRY = COUNTRY_ISO)
                     , by = dplyr::join_by(ADEP)) |>
    dplyr::left_join(.apts_meta |>
                       dplyr::select(ADES = ICAO, DES_CNTRY = COUNTRY_ISO)
                     , by = dplyr::join_by(ADES))

  return(this_mvts)
}

eur_mvts <- eur_mvts |> dplyr::bind_rows() |> prep_eur_network_data()
  # all 2024 flights within EUR airspace ---------------------------
#  eur_mvts |> arrow::write_parquet(
#          "./data-prep-eur/EUR-network-connectivity-analytic-2024.parquet"
#)
  # write out study airport data -----------------------------------
# eur_mvts |>
#   filter(ADEP %in% eur_apts | ADES %in% eur_apts) |>
#   arrow::write_parquet(
#          "./data/EUR-network-connectivity-analytic-2024.parquet")
```

Prepare airport lookup table based on OurAirports.
Patch for missing ICAO location indicators

```r
oa_url <- "https://raw.githubusercontent.com/davidmegginson/ourairports-data/refs/heads/main/
oa <- readr::read_csv(oa_url, show_col_types = F)
oa <- oa |>
  dplyr::filter(!is.na(icao_code)) |>
  dplyr::select(ICAO = icao_code, NAME = name, CNTRY_ISO = iso_country)

# patch oa
patch_oa <- tibble::tribble(
  ~ ICAO, ~ NAME, ~ CNTRY_ISO
  ,"SBCO", "tbd" ,"BR"
```

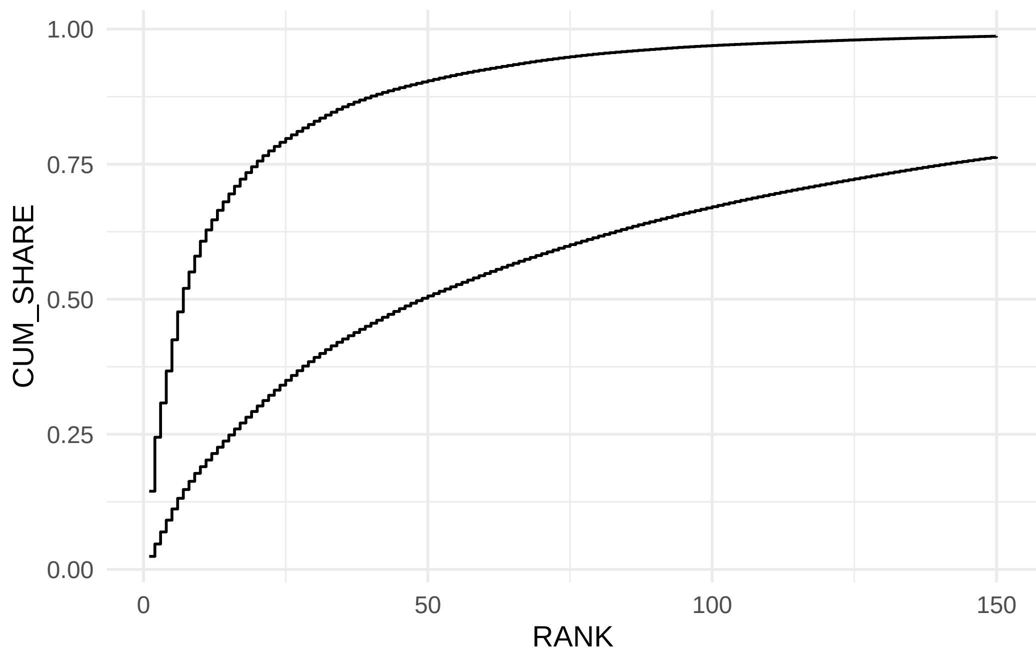
```
  ,"SBJD", "tbd", "BR"
  ,"SBJI", "tbd", "BR"
  ,"SBMI", "tbd", "BR"
 ,"SBMT" , "tbd", "BR"
 ,"SBPO", "tbd", "BR"
 ,"SBRD" , "tbd", "BR"
 ,"SBSG" , "tbd", "BR"
 ,"SBSI", "tbd", "BR"
 ,"SBSO", "tbd", "BR"
,"SBUY", "tbd", "BR"
,"SBVC" , "tbd", "BR"
,'GQNN', NA , "MR"    # Maritania
,'SDAG', NA ,'BR'
,'SDIY', NA ,'BR'
,'SDJV', NA ,'BR'
,'SDLO', NA ,'BR'
,'SDVZ', NA ,'BR'
,'SIFC', NA ,'BR'
,'SIRI', NA ,'BR'
,'SIUA', NA ,'BR'
,'SIXE', NA ,'BR'
,'SJZA', NA ,'BR'
,'SN6L', NA ,'BR'
,'SNAB', NA ,'BR'
,'SNBA', NA ,'BR'
,'SNCL', NA ,'BR'
,"SNEB", NA, "BR"
,"SNIG" , NA, "BR"
,"SNJM"  , NA, "BR"
,"SNLN" , NA, "BR"
,"SNRJ"  , NA, "BR"
,"SNSM"  , NA, "BR"
,"SNSS"  , NA, "BR"
,"SNTS"  , NA, "BR"
,"SNWS"  , NA, "BR"
,"SNZR"  , NA, "BR"
,"SSBN"  , NA, "BR"
,"SSGG"  , NA, "BR"
,"SSQO"  , NA, "BR"
,"SSRS"  , NA, "BR"
,"SSVL"  , NA, "BR"
,"SWBE"  , NA, "BR"
```

```
,"SWKQ"  , NA, "BR"
,"SWYN"  , NA, "BR"
,"SYEC"  , NA, "GN"
)
oa <- oa |> dplyr::bind_rows(patch_oa)
```

```
calc_summary_routes_dep <- function(.mvts){
  this_summary <- .mvts |>
    dplyr::group_by(ADEP) |>
    dplyr::reframe(
       DEPS    = dplyr::n()
      ,NBR_DES = dplyr::n_distinct(ADES)) |>
    dplyr::arrange(dplyr::desc(NBR_DES))

  return(this_summary)
}

if(!exists("icao_iso_other")){
  icao_iso_other <- readr::read_csv("./data/country-icao-iso-etc.csv", show_col_types = F)
}

bra_deps_routes <- bra_mvts |> calc_summary_routes_dep() |> mutate(REGION = "BRA")
eur_deps_routes <- eur_mvts |> bind_rows() |>
  calc_summary_routes_dep() |>
  left_join(oa |> select(ADEP = ICAO, CNTRY_ISO), by = join_by(ADEP)) |>
  #------- patch unknown airports ------------------
  mutate(CNTRY_ISO = case_when(
     grepl(pattern = "^EB", x = ADEP) ~ "BE"
    ,grepl(pattern = "^E[DT]", x = ADEP) ~ "DE"
    ,grepl(pattern = "^EG", x = ADEP) ~ "GB"
    ,grepl(pattern = "^EH", x = ADEP) ~ "NL"
    ,grepl(pattern = "^EI", x = ADEP) ~ "IE"
    ,grepl(pattern = "^EN", x = ADEP) ~ "NO"
    ,grepl(pattern = "^EP", x = ADEP) ~ "PL"
    ,grepl(pattern = "^LE", x = ADEP) ~ "ES"
    ,grepl(pattern = "^LF", x = ADEP) ~ "FR"
    ,grepl(pattern = "^LG", x = ADEP) ~ "GR"
    ,grepl(pattern = "^LH", x = ADEP) ~ "HU"
    ,grepl(pattern = "^LI", x = ADEP) ~ "IT"
    ,grepl(pattern = "^LK", x = ADEP) ~ "CZ"
    ,grepl(pattern = "^LL", x = ADEP) ~ "IL"    # Israel
    ,grepl(pattern = "^LO", x = ADEP) ~ "AT"
```

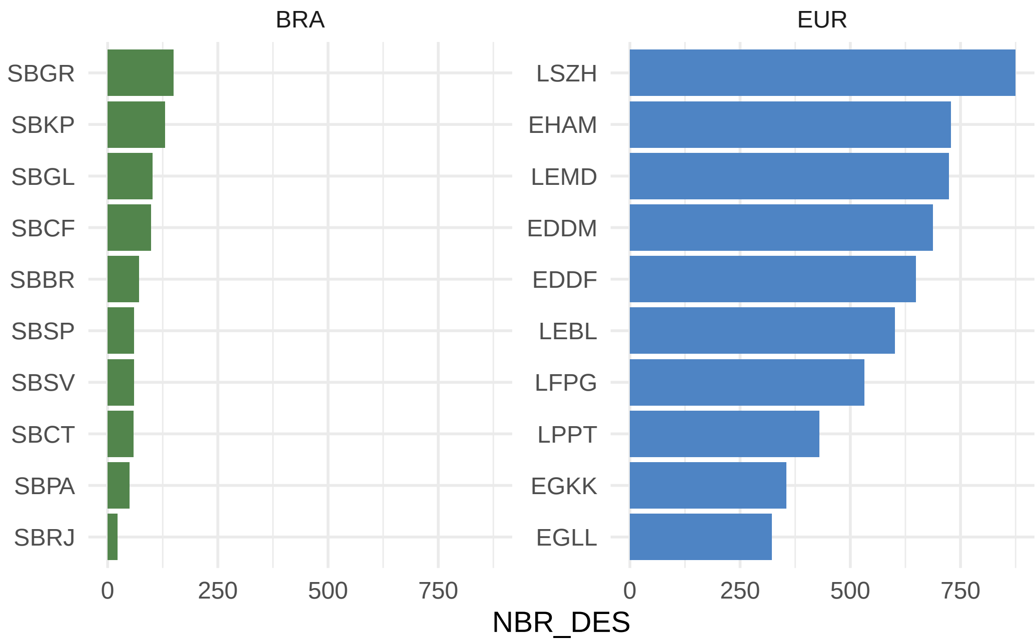
```r
    ,grepl(pattern = "^LR", x = ADEP) ~ "RO"
    #---------- other ADEPs ----------------------
    ,grepl(pattern = "^GM", x = ADEP) ~ "MA"    # Morocco
    ,.default = CNTRY_ISO
  )) |>
  filter(CNTRY_ISO %in% (icao_iso_other |> filter(eurocontrol_pru %in% "Eurocontrol") |> pul
  mutate(REGION = "EUR")

# TODO - save as interim data and work from there
comb_deps_routes <- bind_rows(bra_deps_routes, eur_deps_routes) |>
  group_by(REGION) |>
  arrange(desc(DEPS), .by_group = TRUE) |>
  mutate(RANK = 1:n(), SHARE = DEPS / sum(DEPS), CUM_SHARE = cumsum(SHARE)
         ,IN_STUDY = ADEP %in% c(bra_apts, eur_apts)) |>
  ungroup()

#comb_deps_routes |> arrow::write_parquet("./data/BRA-EUR-dep-counts-2024.parquet")
```

```r
if(exists("comb_deps_routes")){
  comb_deps_routes <- arrow::read_parquet("./data/BRA-EUR-dep-counts-2024.parquet")
}

#comb_deps_routes |> group_by(REGION) |> mutate(RANK = RANK / max(RANK)) |>
comb_deps_routes |> filter(RANK <= 150) |>
  ggplot() +
  geom_step(aes(x = RANK, y = CUM_SHARE, group = REGION), direction = "hv")
```




```
bind_rows(
   bra_deps_routes |> filter(ADEP %in% bra_apts) |> mutate(REGION = "BRA")
  ,eur_deps_routes |> filter(ADEP %in% eur_apts) |> mutate(REGION = "EUR")
  ) |>
  mutate(ADEP = fct_infreq(ADEP, NBR_DES) |> fct_rev()) |>
  ggplot() +
  geom_col(aes(x = NBR_DES, y = ADEP, fill = REGION)) +
  scale_fill_manual(values = bra_eur_colours) +
  facet_wrap(.~ REGION, scales = "free_y") +
  labs(y = element_blank()) +
  theme(legend.position = "none")
```

```
Warning: `label` cannot be a <ggplot2::element_blank> object.
```




```
get_spread_of_destinations <- function(.deps){
  this_spread <- .deps |>
    dplyr::group_by(ADEP, DES_CNTRY) |>
    dplyr::reframe(
            DEPS = dplyr::n()
          , DEST = dplyr::n_distinct(ADES)) |>
    dplyr::group_by(ADEP) |>
    dplyr::mutate(
           SHARE_DEP      = DEPS / sum(DEPS)
         , TOP_DEST_SHARE = DEST / sum(DEST)
         ) |>
    dplyr::arrange(desc(DEPS), .by_group = TRUE) |>
    dplyr::ungroup()

  return(this_spread)
}

#------------------- BRA ---------------------------------------------
bra_dests <- bra_mvts |>
  filter(ADEP %in% bra_apts) |>
  get_spread_of_destinations()

#------------------ EUR --------------------------------------------------
```

```r
eur_mvts_study <- arrow::read_parquet("./data/EUR-network-connectivity-analytic-2024.parquet"
ectrl_ms       <- readxl::read_xlsx("./data/ECTRL-ICAOEUR-state-memberships.xlsx") |>   filte
  mutate(REGION = "EUR")

#------------ STATE and ISO CODES ----------------------------------------
global_iso2    <- ISOcodes::ISO_3166_1 |>
  tibble() |>
  select(STATE_ISO = Alpha_2, STATE = Name)

patch_global <- tibble::tribble(
  ~ STATE_ISO, ~STATE,
  "AN", "Netherlands Antilles",
  "XK", "Kosovo"      # need to think how to add to EUR region
)
global_iso2 <- global_iso2 |> bind_rows(patch_global)

# world wide codes - we may need this for generalising BRA data
# constructed wiht ChatGPT - not validated !!!!
icao_iso2 <- read_csv("./data/country-icao-iso-territory.csv")
```

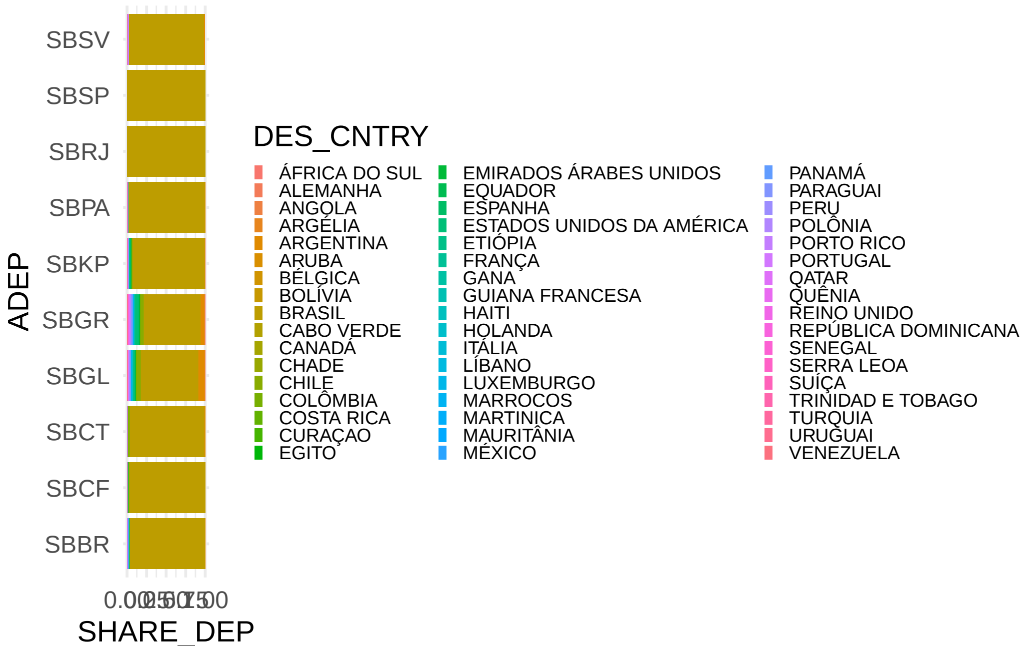
```
Rows: 218 Columns: 4
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (4): ICAO Region, Country/Territory Name, ISO Alpha-2 Code, Sovereign State

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
oa_url <- "https://raw.githubusercontent.com/davidmegginson/ourairports-data/refs/heads/main/
oa <- readr::read_csv(oa_url, show_col_types = F)
oa <- oa |>
  dplyr::filter(!is.na(icao_code)) |>
  dplyr::select(ICAO = icao_code, NAME = name, CNTRY_ISO = iso_country)

# patch oa
patch_oa <- tibble::tribble(
  ~ ICAO, ~ NAME, ~ CNTRY_ISO
  ,"SBCO", "tbd" ,"BR"
  ,"SBJD", "tbd", "BR"
  ,"SBJI", "tbd", "BR"
  ,"SBMI", "tbd", "BR"
```

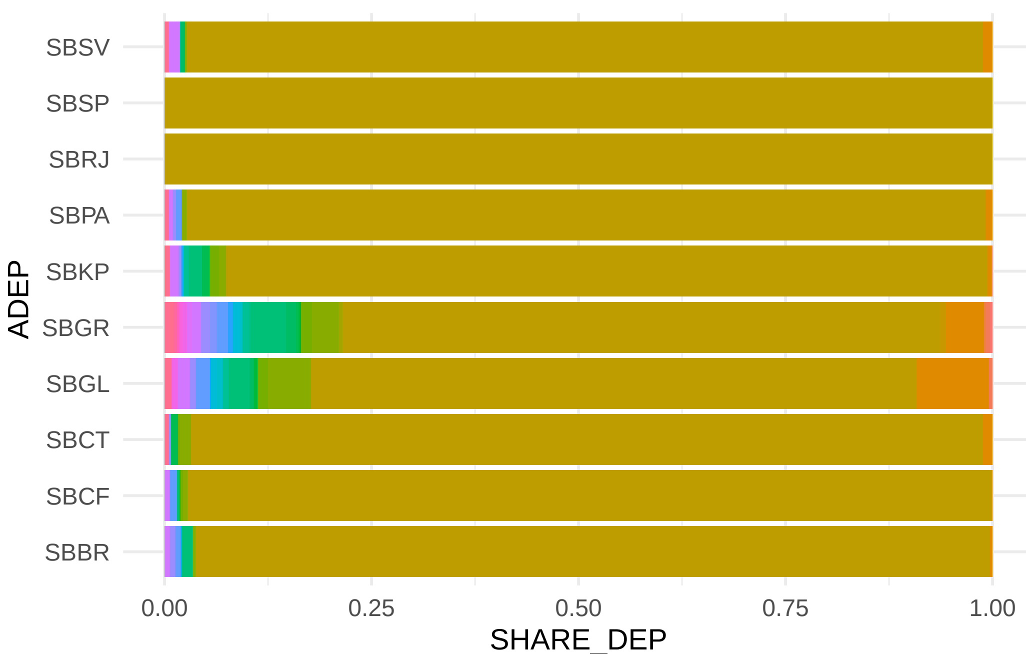
```
 ,"SBMT" , "tbd", "BR"
 ,"SBPO", "tbd", "BR"
 ,"SBRD" , "tbd", "BR"
 ,"SBSG" , "tbd", "BR"
 ,"SBSI", "tbd", "BR"
 ,"SBSO", "tbd", "BR"
,"SBUY", "tbd", "BR"
,"SBVC" , "tbd", "BR"
,'GQNN', NA , "MR"    # Maritania
,'SDAG', NA ,'BR'
,'SDIY', NA ,'BR'
,'SDJV', NA ,'BR'
,'SDLO', NA ,'BR'
,'SDVZ', NA ,'BR'
,'SIFC', NA ,'BR'
,'SIRI', NA ,'BR'
,'SIUA', NA ,'BR'
,'SIXE', NA ,'BR'
,'SJZA', NA ,'BR'
,'SN6L', NA ,'BR'
,'SNAB', NA ,'BR'
,'SNBA', NA ,'BR'
,'SNCL', NA ,'BR'
,"SNEB", NA, "BR"
,"SNIG" , NA, "BR"
,"SNJM"  , NA, "BR"
,"SNLN" , NA, "BR"
,"SNRJ"  , NA, "BR"
,"SNSM"  , NA, "BR"
,"SNSS"  , NA, "BR"
,"SNTS"  , NA, "BR"
,"SNWS"  , NA, "BR"
,"SNZR"  , NA, "BR"
,"SSBN"  , NA, "BR"
,"SSGG"  , NA, "BR"
,"SSQO"  , NA, "BR"
,"SSRS"  , NA, "BR"
,"SSVL"  , NA, "BR"
,"SWBE"  , NA, "BR"
,"SWKQ"  , NA, "BR"
,"SWYN"  , NA, "BR"
,"SYEC"  , NA, "GN"
```

```
)
oa <- oa |> dplyr::bind_rows(patch_oa)

# icao_iso_other <- countrycode::codelist |>
#   select(iso2c, country.name.en, continent, eurocontrol_statfor, eurocontrol_pru, icao.regi
# icao_iso_other |> readr::write_csv("./data/country-icao-iso-etc.csv")
icao_iso_other <- readr::read_csv("./data/country-icao-iso-etc.csv", show_col_types = F)

# to-do patch
#   iso2c country.name.en continent eurocontrol_statfor eurocontrol_pru icao.region
#   <chr> <chr>           <chr>     <chr>               <chr>           <chr>
# 1 SN    Senegal         Africa    NA                  NA              AFI


#------------------------------------------------------ END ISO ----------

eur_dests <- eur_mvts_study |>
  filter(ADEP %in% eur_apts) |>
  get_spread_of_destinations() |>
  #---------- append ECTRL / EUR ---------
  left_join(global_iso2 |>
              rename(DES_CNTRY = STATE_ISO), by = join_by(DES_CNTRY)
            ) |>
  left_join(ectrl_ms |> select(-STATE) |>
              rename(DES_CNTRY = STATE_ISO) , by = join_by(DES_CNTRY)
            )
```

```
p <- bra_dests |>
  ggplot() + geom_col(aes(x = SHARE_DEP, y = ADEP, fill = DES_CNTRY)) +
  theme(
    legend.text = element_text(size = 7),
    legend.key.size = unit(0.3, "lines")
  )
p
```




```r
p + theme(legend.position = "none")
```



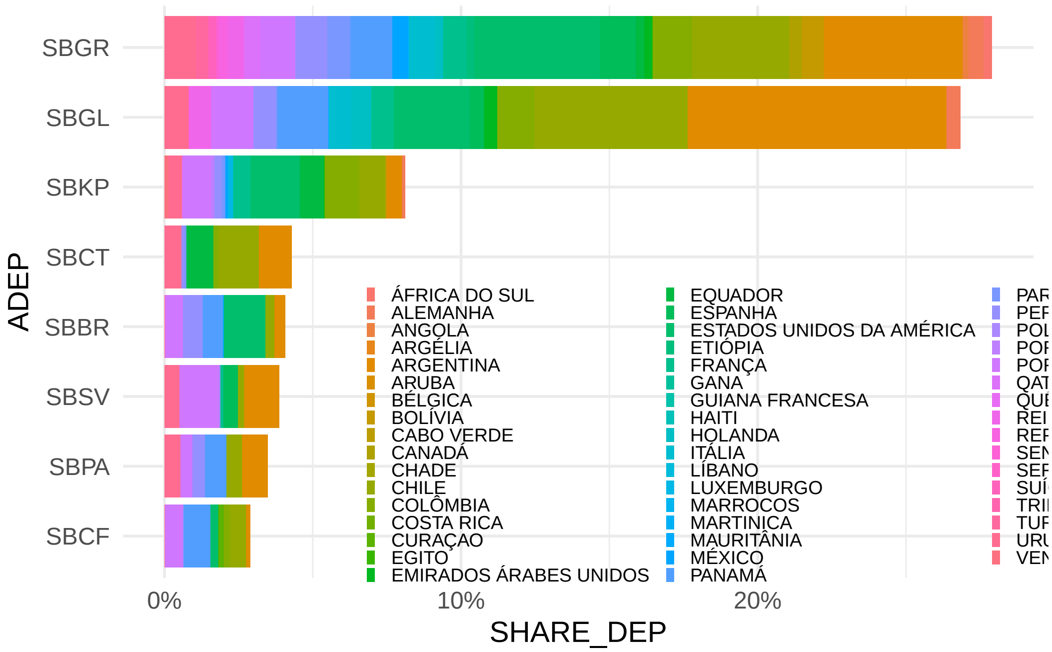
international traffic

```r
bra_dests |> filter(DES_CNTRY != "BRASIL") |>
  mutate(ADEP = fct_infreq(ADEP, SHARE_DEP) |> fct_rev())
```

```
# A tibble: 145 x 6
   ADEP  DES_CNTRY                  DEPS  DEST SHARE_DEP TOP_DEST_SHARE
   <fct> <chr>                     <int> <int>     <dbl>          <dbl>
 1 SBBR  ESTADOS UNIDOS DA AMÉRICA   770     3 0.0141            0.0423
 2 SBBR  PANAMÁ                      372     1 0.00682           0.0141
 3 SBBR  PERU                        365     1 0.00670           0.0141
 4 SBBR  PORTUGAL                    326     1 0.00598           0.0141
 5 SBBR  ARGENTINA                   194     2 0.00356           0.0282
 6 SBBR  CHILE                       165     2 0.00303           0.0282
 7 SBBR  URUGUAI                       8     1 0.000147          0.0141
 8 SBBR  MÉXICO                        7     1 0.000128          0.0141
 9 SBBR  VENEZUELA                     3     1 0.0000550         0.0141
10 SBBR  MARROCOS                      2     1 0.0000367         0.0141
# i 135 more rows
```

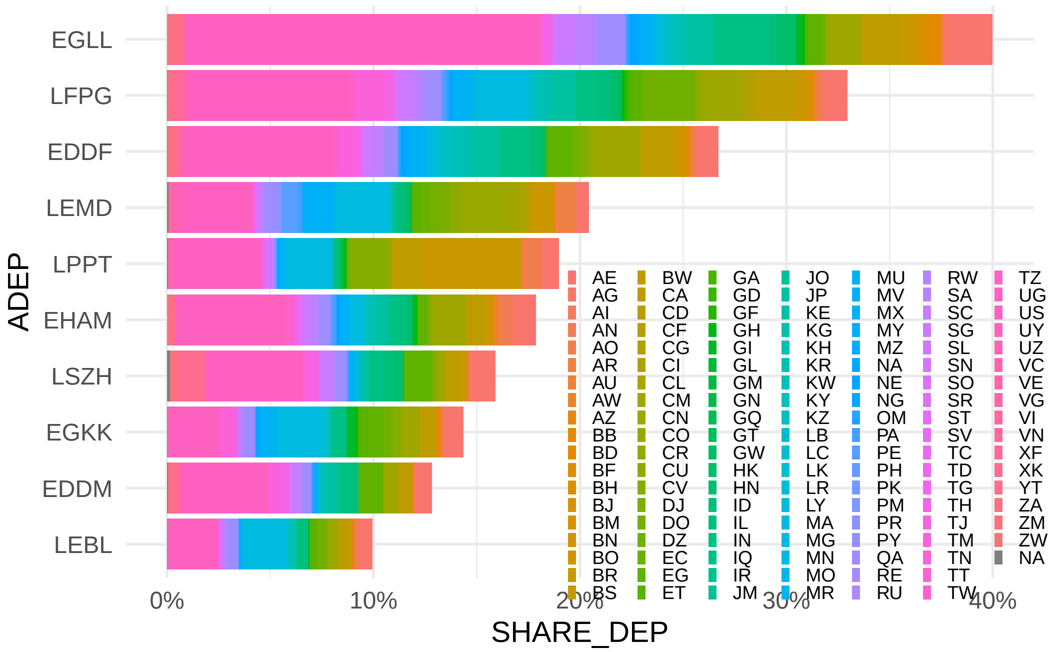
```r
# DES_CNTRY ~ PAIS_DESTINO

bra_dests |> filter(DES_CNTRY != "BRASIL") |>
  mutate(ADEP = fct_infreq(ADEP, SHARE_DEP) |> fct_rev()) |>

  ggplot() + geom_col(aes(x = SHARE_DEP, y = ADEP, fill = DES_CNTRY)) +
  scale_x_continuous(labels = scales::percent_format()) +
  labs(fill = element_blank()) +
  theme(
    legend.text = element_text(size = 7),
    legend.key.size = unit(0.3, "lines"),
    legend.position = "inside", legend.position.inside = c(0.75,0.25)
  )
```

```
Warning: `label` cannot be a <ggplot2::element_blank> object.
```




```r
eur_dests |>
  filter(is.na(REGION)) |>
  mutate(ADEP = fct_infreq(ADEP, SHARE_DEP) |> fct_rev()) |>

  ggplot() + geom_col(aes(x = SHARE_DEP, y = ADEP, fill = DES_CNTRY)) +
  scale_x_continuous(labels = scales::percent_format()) +
  labs(fill = element_blank()) +
  theme(
    legend.text = element_text(size = 7),
    legend.key.size = unit(0.3, "lines"),
    legend.position = "inside", legend.position.inside = c(0.75,0.25)
  )
```

```
Warning: `label` cannot be a <ggplot2::element_blank> object.
```



Appendix 1 to the Directives to Regional Air Navigation Meetings and Rules of Procedure for their Conduct (Doc 8144-AN/874), comprises the following regions:

- Africa-Indian Ocean Region (AFI);
- Asia Region (ASIA);
- Caribbean Region (CAR);
- European Region (EUR);
- Middle East Region (MID);
- North American Region (NAM);
- North Atlantic Region (NAT);
- Pacific Region (PAC);
- South American Region (SAM).

```
eur_world_shares <- eur_dests |>
  filter(is.na(REGION)) |>
  mutate(ADEP = fct_infreq(ADEP, SHARE_DEP) |> fct_rev()) |>
  left_join(icao_iso2 |> rename(DES_CNTRY = `ISO Alpha-2 Code`, ICAO_REG = `ICAO Region`) |>
              select(DES_CNTRY, ICAO_REG)) |>
  # aggregate by ICAO Region
  group_by(ICAO_REG) |>
  reframe(DEPS = sum(DEPS), DEST = sum(DEST)) |>
  mutate(REG_SHARES = DEPS / sum(DEPS))
```

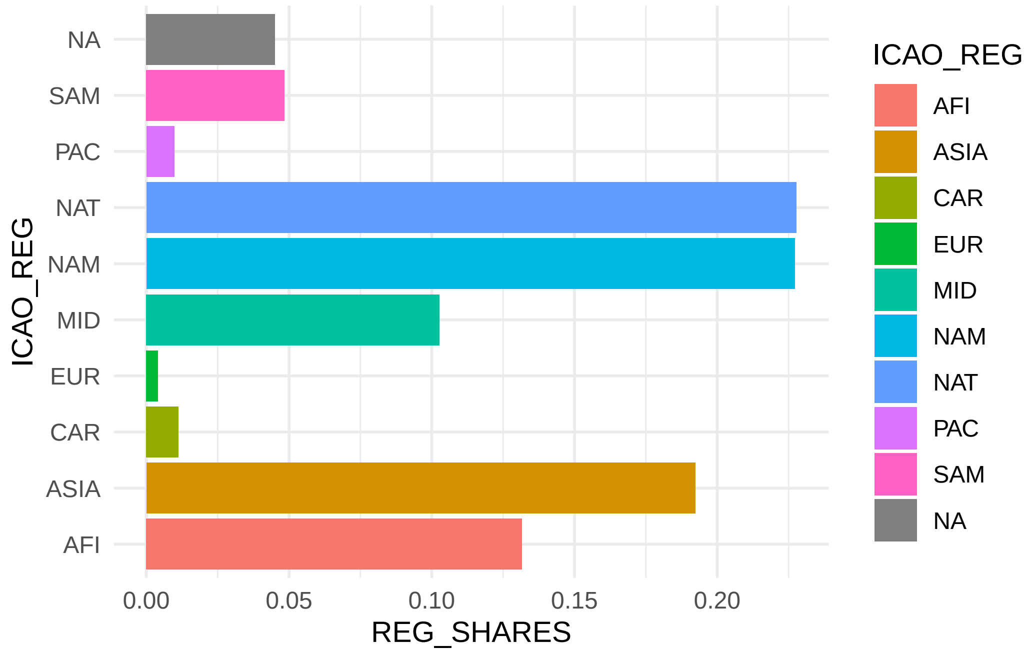
```
Joining with `by = join_by(DES_CNTRY)`
```

```
Warning in left_join(mutate(filter(eur_dests, is.na(REGION)), ADEP = fct_rev(fct_infreq(ADEP
to-many relationship between `x` and `y`.
i Row 1 of `x` matches multiple rows in `y`.
i Row 175 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```

```r
eur_world_shares
```

```
# A tibble: 10 x 4
   ICAO_REG   DEPS  DEST REG_SHARES
   <chr>     <int> <int>      <dbl>
 1 AFI       78359   664    0.132
 2 ASIA     114511   511    0.192
 3 CAR        6694    77    0.0112
 4 EUR        2421     8    0.00407
 5 MID       61143   253    0.103
 6 NAM      135251   648    0.227
 7 NAT      135579   656    0.228
 8 PAC        5818    11    0.00977
 9 SAM       28849   110    0.0484
10 <NA>      26832   131    0.0451
```

```r
eur_world_shares |>
  ggplot() +
  geom_col(aes(x = REG_SHARES, y = ICAO_REG, fill = ICAO_REG))
```



EUR ~ non-ECTRL Europe (probably need to take this out)

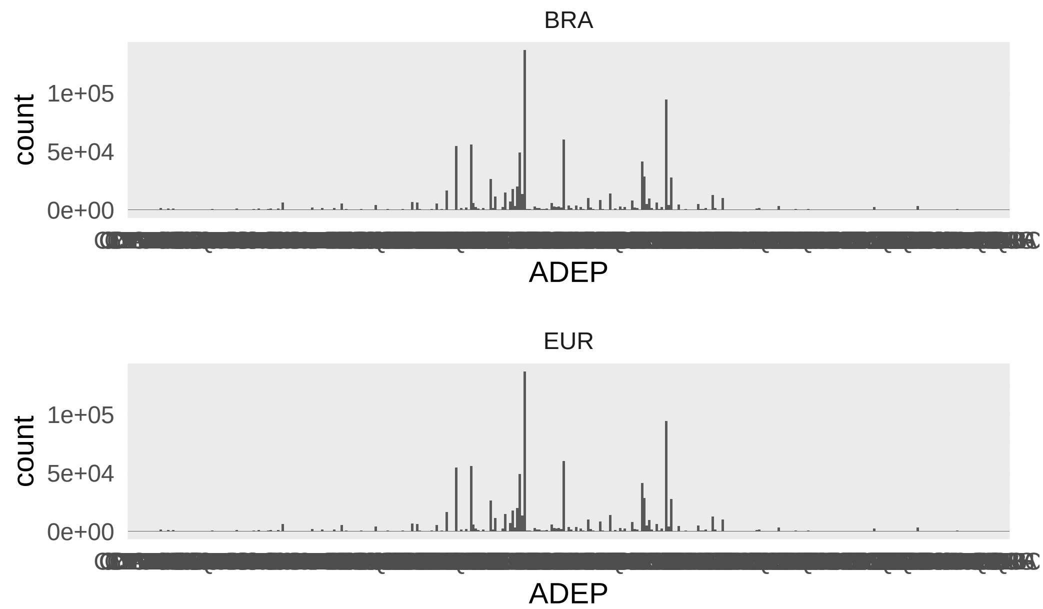
Workflow Demo - Getting from Code to Functions

```
calc_adep_counts <- function(.ds){
  my_count <- .ds |>
  group_by(ADEP, DEP_CNTRY) |>  # Group by ADEP and PAIS_ORIGEM
  summarise(count = n(), .groups = "drop") |>
  arrange(desc(count))

  return(my_count)
}



bra_mvts |> calc_adep_counts()
```

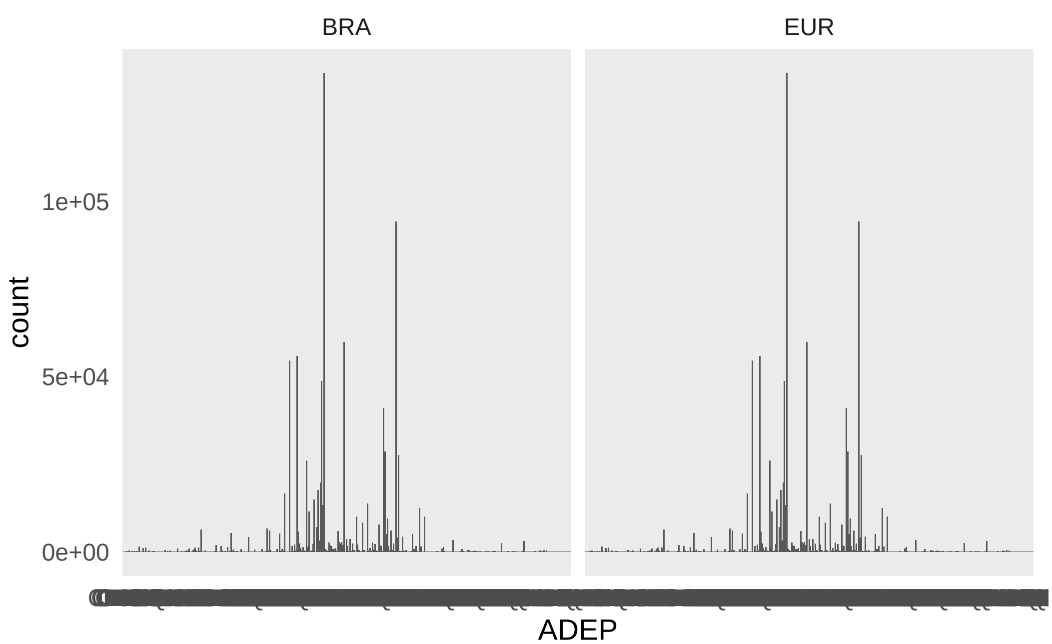
```
# A tibble: 361 x 3
   ADEP  DEP_CNTRY  count
   <chr> <chr>      <int>
```

```
 1 SBGR  BRASIL    136561
 2 SBSP  BRASIL     94262
 3 SBKP  BRASIL     59915
 4 SBCF  BRASIL     55898
 5 SBBR  BRASIL     54509
 6 SBGL  BRASIL     48774
 7 SBRF  BRASIL     41092
 8 SBRJ  BRASIL     28620
 9 SBSV  BRASIL     27645
10 SBCT  BRASIL     26098
# i 351 more rows
```

```r
plot_barchart <- function(.ds){

  p <- .ds |>
  ggplot() +
  geom_col(aes(x = ADEP, y = count)) +
    facet_wrap(.~ REGION)

  return(p)
}

bra_demo <- bra_mvts |> calc_adep_counts() |> mutate(REGION = "BRA")
eur_demo <- bra_mvts |> calc_adep_counts() |> mutate(REGION = "EUR")

p1 <- bra_demo |> plot_barchart()
p2 <- eur_demo |> plot_barchart()

p1 / p2
```




```
bind_rows(bra_demo, eur_demo) |> plot_barchart()
```



The maximum departures seen at Brazilian airport account for 136561.

Prepare analytic data for study

[1] "ADEP" "ADES" "FLTID" "OPR" "REG" "TYPE" "FLT_DUR"
[8] "DATE" "SAM_ID" "DEP_CNTRY" "DES_CNTRY"

```
bra_mvts <- read_csv("./data_proc/flights_ANAC_2024.csv"
                   #   , col_select = dplyr::any_of(cols)
                     , show_col_types = FALSE
                     )
```

```
New names:
* `` -> `...1`
```

```
trim_ANAC <- function(.anac){
  trimmed_anac <- .anac |>
    dplyr::select(
       ADEP     = sg_icao_origem
      ,ADES     = sg_icao_destino
      ,FLT_NBR = nr_voo
      ,OPR  = sg_empresa_icao           # aircraft operator (ICAO)
      ,REG  = ds_matricula
      ,TYPE = sg_equipamento_icao       # aircraft type (ICAO)
      ,AOBT     = hr_partida_real
      ,AIBT     = hr_chegada_real
      ,FLT_DUR = nr_horas_voadas        # flight duration
      ,DATE     = dt_referencia
      ,id_basica                        # ANAC ID
      ,DEP_CNTRY_NAME = nm_pais_origem  # departing country
      ,DES_CNTRY_NAME = nm_pais_destino # destination country
    )
  return(trimmed_anac)
}

prep_bra_connectivity <- function(.trimmed_anac){
  bra_analytic <- .trimmed_anac |>
    dplyr::mutate(
       FLTID = paste0(OPR, FLT_NBR)
    )
 return(bra_analytic)
}
```

```r
append_iso_adep_ades <- function(
      .flt_adep_ades
    , .iso_look_up = oa |> select(ICAO, CNTRY_ISO)
    ){
  this_flt <- .flt_adep_ades |>
    dplyr::left_join(
       .iso_look_up |> dplyr::rename(ADEP = ICAO, DEP_CNTRY = CNTRY_ISO)
      , by = dplyr::join_by(ADEP)
      ) |>
    dplyr::left_join(
      .iso_look_up |> dplyr::rename(ADES = ICAO, DES_CNTRY = CNTRY_ISO)
      , by = dplyr::join_by(ADES))

  return(this_flt)
}

append_ades_regions <- function(
     .flt_ades_iso2
    ,.region_lookup = icao_iso_other |>
      filter(!is.na(iso2c)) |>
      select(  DES_CNTRY = iso2c
             , DES_ICAO  = icao.region
             , DES_STATF = eurocontrol_statfor
             , DES_PRU = eurocontrol_pru)
){
  this_flt <- .flt_ades_iso2 |>
    dplyr::left_join(.region_lookup, by = dplyr::join_by(DES_CNTRY))

  return(this_flt)
}

append_adep_regions <- function(
     .flt_adep_iso2
    ,.region_lookup = icao_iso_other |>
      filter(!is.na(iso2c)) |>
      select(  DEP_CNTRY = iso2c
             , DEP_ICAO  = icao.region
             , DEP_STATF = eurocontrol_statfor
             , DEP_PRU = eurocontrol_pru)
){
  this_flt <- .flt_adep_iso2 |>
    dplyr::left_join(.region_lookup, by = dplyr::join_by(DEP_CNTRY))
```

```
  return(this_flt)
}
```

```
eur_mvts |> append_adep_regions() |>
  append_ades_regions() |> filter(year(DATE) == 2024)
```

```
# A tibble: 10,211,794 x 17
   ADEP  ADES  FLTID   OPR   REG    TYPE  FLT_DUR DATE          SAM_ID DEP_CNTRY
   <chr> <chr> <chr>   <chr> <chr>  <chr>   <int> <date>         <dbl> <chr>
 1 OMDB  UUDD  UAE129  UAE   A6ECX  B77W      297 2024-01-01 268552706 AE
 2 MROC  LEMD  IBE6314 IBE   ECMOU  A332      553 2024-01-01 268552700 CR
 3 SEQM  LEMD  IBE6454 IBE   ECMXV  A359      569 2024-01-01 268552704 EC
 4 KORD  EDDM  UAL953  UAL   N12020 B78X      480 2024-01-01 268552701 US
 5 KORD  EGLL  AAL90   AAL   N826AN B789      421 2024-01-01 268552702 US
 6 KJFK  EHAM  DAL48   DAL   N814NW A333      387 2024-01-01 268552705 US
 7 KPHL  LSZH  AAL92   AAL   N881BK B788      411 2024-01-01 268552703 US
 8 KJFK  EGLL  VIR46K  VIR   GVTOM  A339      376 2024-01-01 268552707 US
 9 KORD  LFPG  AAL150  AAL   N816AA B788      442 2024-01-01 268552708 US
10 GMMN  DGAA  RAM515F RAM   CNROR  B738      236 2024-01-01 268552716 MA
# i 10,211,784 more rows
# i 7 more variables: DES_CNTRY <chr>, DEP_ICAO <chr>, DEP_STATF <chr>,
#   DEP_PRU <chr>, DES_ICAO <chr>, DES_STATF <chr>, DES_PRU <chr>
```

```
# eur_mvts |>
#   arrow::write_parquet(sink = here::here("data_proc","EUR-network-connectivity-2024.parque

bra_mvts <- bra_mvts |> trim_ANAC() |> append_iso_adep_ades() |>
  append_adep_regions() |>
  append_ades_regions()
```

```
Warning in dplyr::left_join(.flt_adep_ades, dplyr::rename(.iso_look_up, : Detected an unexpe
to-many relationship between `x` and `y`.
i Row 3021 of `x` matches multiple rows in `y`.
i Row 7117 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```

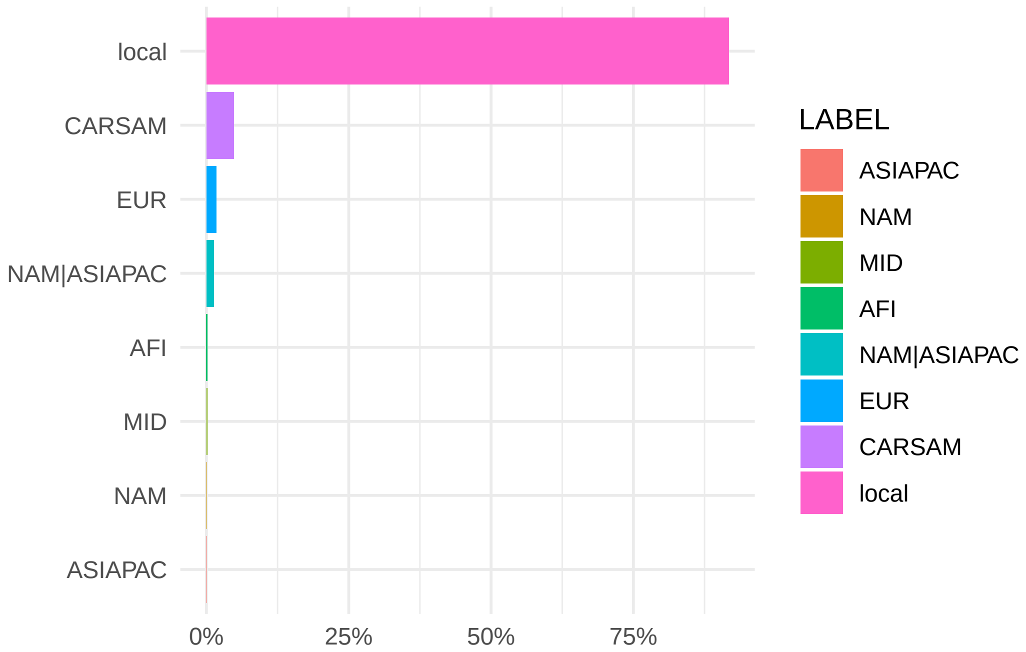
```
Warning in dplyr::left_join(dplyr::left_join(.flt_adep_ades, dplyr::rename(.iso_look_up, : De
to-many relationship between `x` and `y`.
i Row 2946 of `x` matches multiple rows in `y`.
```

```
i Row 6991 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```

```
# bra_mvts |> arrow::write_parquet(sink = here::here("data_proc","BRA-network-connectivity-2(
```

```
bra_reg_tfc <- bra_mvts |>
  mutate(DOMESTIC = DES_CNTRY == "BR") |>
  group_by(DES_ICAO, DOMESTIC) |>
  reframe(DEPS = n())

bra_reg_tfc |> mutate(LABEL = ifelse(!DOMESTIC, DES_ICAO, "local"), SHARE = DEPS / sum(DEPS)
```

```
Warning: `label` cannot be a <ggplot2::element_blank> object.
`label` cannot be a <ggplot2::element_blank> object.
```




```
library(ComplexUpset)

dests_bra_eur <- bra_mvts |>
  filter(DES_PRU == "Eurocontrol", DEP_CNTRY == "BR") |>
  group_by(ADEP, ADES) |>
```

```r
  reframe(DEPS = n()) |>
  arrange(desc(DEPS)) |>
  mutate(SHARE = DEPS / sum(DEPS), DUMMY = 1
         ,CUM_SHARE = cumsum(SHARE))

tmp <- bra_mvts |>
  filter(DEP_CNTRY == "BR", DES_PRU == "Eurocontrol") |>
  group_by(ADEP, ADES, DES_PRU, DEP_CNTRY) |>
  reframe(DEPS = n()) |>
  arrange(desc(DEPS)) |>
  mutate(SHARE = DEPS / sum(DEPS), CUM_SHARE = cumsum(SHARE)) |>
  mutate(IS_STUDY = ADES %in% eur_apts) |>
  pivot_wider(names_from = ADES, values_from = IS_STUDY)

tmp <- tmp |> filter(CUM_SHARE <= 0.8)
tiks <- names(tmp)[-c(1:4)]
# THROWS ERROR tmp |> upset(tiks)
```

another cool and long airport lookup

```r
ip2_url <- "https://github.com/ip2location/ip2location-iata-icao/blob/master/iata-icao.csv"

# github differentiates between "display" and "raw" version
# we download raw versions!
ip2_url <- "https://raw.githubusercontent.com/ip2location/ip2location-iata-icao/refs/heads/ma


icao_ip2 <- readr::read_csv(ip2_url)
```

```
Rows: 9160 Columns: 7
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (5): country_code, region_name, iata, icao, airport
dbl (2): latitude, longitude

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
bra_mvts <- arrow::read_parquet("data_proc/BRA-network-connectivity-2024.parquet")
eur_mvts <- arrow::read_parquet("data_proc/EUR-network-connectivity-2024.parquet")
```

international traffic

```
bra_mvts_intnl <- bra_mvts |>
  filter(DEP_CNTRY == "BR") |>
  mutate(
     TFC_TYPE = ifelse(DEP_CNTRY == "BR" & DES_CNTRY == "BR", "regional", "international")
    ,IS_STUDY = ADEP %in% bra_apts) |>
    group_by(DEP_CNTRY, IS_STUDY, DES_CNTRY, TFC_TYPE, DES_PRU) |> reframe(DEPS = n())  |> a

# bra_mvts_intnl |> mutate(YEAR = 2024) |> write_csv("./data/BRA-network-conns-intnl.csv")

eur_mvts_intnl <- eur_mvts |>
  filter(DEP_PRU == "Eurocontrol") |>
  mutate(TFC_TYPE = ifelse(DEP_PRU == "Eurocontrol" & DES_PRU == "Eurocontrol", "regional","
    group_by(DEP_CNTRY, IS_STUDY, DES_CNTRY, TFC_TYPE, DES_PRU) |>
  reframe(DEPS = n()) |>
  arrange(desc(DEPS))

#eur_mvts_intnl |> mutate(REGION = "EUR", YEAR = 2024) |>
#  write_csv("./data/EUR-network-conns-intnl.csv")
```

upset presentation

for upset plot we need to identify the group membership"S".

```
library(ComplexUpset)

bra_neighbours <- tribble(
  ~ISO2, ~CNTRY_NAME
  , "AR", "Argentina"
  , "CL", "Chile"
  , "CO", "Colombia"
  , "UY", "Uruguay")

bra_upset <- bra_mvts |>
  mutate(ID = row_number(), IS_STUDY = ifelse(ADEP %in% bra_apts, 1, 0)
         , CASE = paste0(DEP_CNTRY,"-",DES_CNTRY,"_",IS_STUDY)
         ,NEIGHBOUR = case_when(
           DES_CNTRY %in% bra_neighbours$ISO2 ~ DES_CNTRY
          , .default = "Other")
         , DUMMY = 1 ) |>

  filter(DES_PRU %in% c("Southern America"), DES_CNTRY != "BR") |>
```

```r
  select(ID,CASE, IS_STUDY, DEP_CNTRY, DES_CNTRY, DES_PRU, DUMMY, NEIGHBOUR) |>
  mutate(  STUDY_APT = ifelse(IS_STUDY, TRUE, FALSE)
         , NON_STUDY_APT = ifelse(!IS_STUDY, TRUE, FALSE)) |>
  pivot_wider(id_cols = c(ID, CASE, DES_CNTRY, DES_PRU, STUDY_APT, NON_STUDY_APT)
              , names_from = NEIGHBOUR
              , values_from = DUMMY, values_fill = 0)

# make nice names
# look up (named) vector
bra_neighbours_vec <- deframe(bra_neighbours)
study_apt_label    <- c(STUDY_APT = "airport in study", NON_STUDY_APT = "non-study airport")
nice_name_vec     <- c(bra_neighbours_vec, study_apt_label)

bra_upset <- bra_upset |>
  rename_with(.fn = ~ nice_name_vec[.x], .cols = any_of(names(nice_name_vec)))

tiks <- names(bra_upset)[-c(1:4)]
#tiks <- c("STUDY_APT", "Argentina", "Chile", "Columbia", "Uruguay", "Other", "NON_STUDY_APT


plot_bra_upset <- bra_upset |> upset(
    intersect = tiks
  , name = "connections"
  , base_annotations = list(
    # fine-tune presentation of Intersection size portion
    'Intersection size'=intersection_size(
            fill = bra_col,
            text_colors = c(
                on_background='black', on_bar='white'
            )
            ,text_mapping = aes(
              label = paste0(
                 !!get_size_mode('exclusive_intersection')
                # ,"\n"
                # ,!!upset_text_percentage()
                )
            )
            , text = list(size = 9 , size.unit = "pt")
        )
    # add ggplot layer for total numbers
        + annotate(
            geom='text', x=Inf, y=Inf,
```

```
            label=paste('total number \nof non-regional \nflights:', nrow(bra_upset)),
            vjust=1, hjust=1
        ) +
      labs(y = "departures")
  ),
  # fine-tune query part
  # queries=list(upset_query(set='airport in study', fill='blue'))
  # remove the sizes/query part ... not sure what it adds
  set_sizes = FALSE
)
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
i The deprecated feature was likely used in the ComplexUpset package.
  Please report the issue at
  <https://github.com/krassowski/complex-upset/issues>.
```

```
Warning: themes$intersections_matrix is not a valid theme.
Please use `theme()` to construct themes.
```

```
Warning: `legend.margin` must be specified using `margin()`
i For the old behavior use `legend.spacing`
```

```
Warning: themes$intersections_matrix is not a valid theme.
Please use `theme()` to construct themes.
```

```
Warning: selected_theme is not a valid theme.
Please use `theme()` to construct themes.
```

```
Warning: `legend.margin` must be specified using `margin()`
i For the old behavior use `legend.spacing`
```

```
Warning: selected_theme is not a valid theme.
Please use `theme()` to construct themes.
```

```
# THROWS ERROR - CHECK plot_bra_upset
```

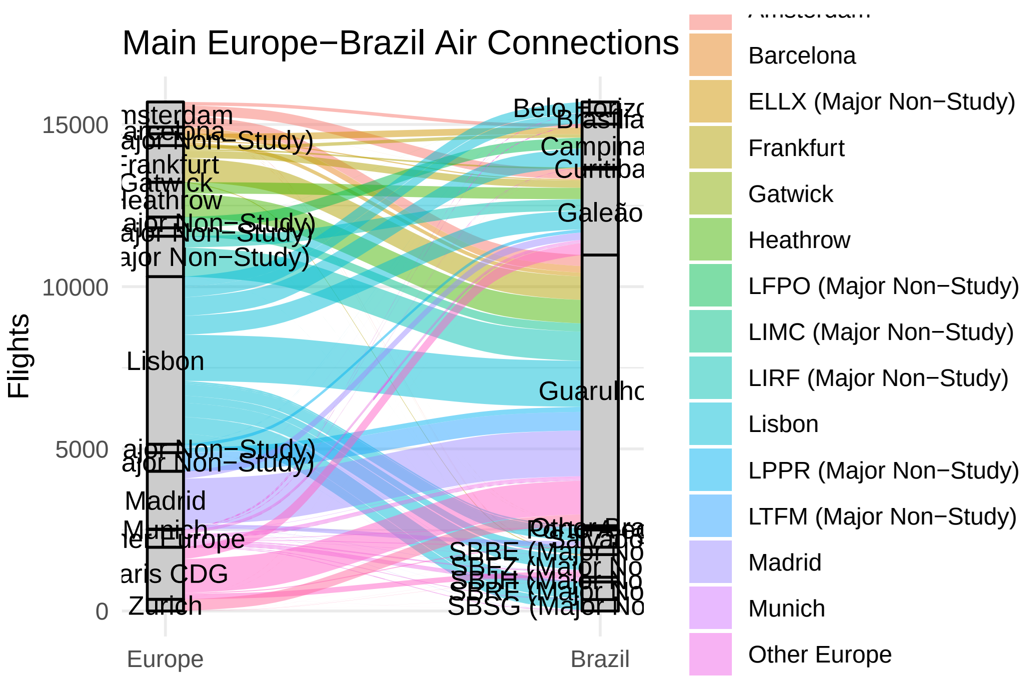
```
# save plot for report
plot_bra_upset |> ggsave(filename = "./figures/BRA-upset-national-connections.png", width = 8
```

BRA-EUR traffic

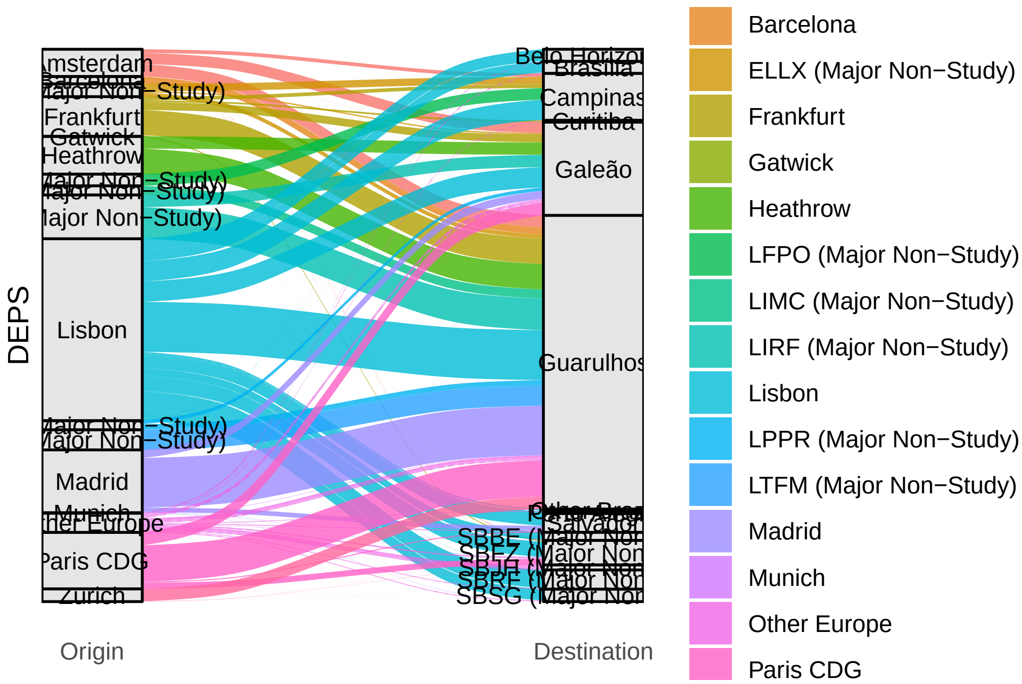
```
study_br <- c("SBBR", "SBCF", "SBCT", "SBGL", "SBGR", "SBKP", "SBPA", "SBRJ", "SBSP", "SBSV")
study_eu <- c("EDDF", "EDDM", "EGKK", "EGLL", "EHAM", "LEBL", "LEMD", "LFPG", "LPPT", "LSZH")

# Named vectors for readable names
br_labels <- c(
  SBBR = "Brasília", SBCF = "Belo Horizonte", SBCT = "Curitiba",
  SBGL = "Galeão", SBGR = "Guarulhos", SBKP = "Campinas", SBPA = "Porto Alegre",
  SBRJ = "Santos Dumont", SBSP = "Congonhas", SBSV = "Salvador"
)

eu_labels <- c(
  EDDF = "Frankfurt", EDDM = "Munich", EGKK = "Gatwick", EGLL = "Heathrow",
  EHAM = "Amsterdam", LEBL = "Barcelona", LEMD = "Madrid", LFPG = "Paris CDG",
  LPPT = "Lisbon", LSZH = "Zurich"
)

# trim connections
eur_bra_conns <- eur_mvts |>
  filter(DEP_PRU == "Eurocontrol", DES_CNTRY == "BR") |>
  group_by(ADEP, ADES) |> reframe(DEPS = n()) |> arrange(desc(DEPS))

conns_threshold <- 90

major_nonstudy_eur <- eur_bra_conns |>
  filter(!(ADEP %in%  study_eu)) |>   # only exclude study airports on the origin side
  group_by(ADEP)  |>
  reframe(TOT_DEPS = sum(DEPS))  |>
  filter(TOT_DEPS >= conns_threshold) %>%
  pull(ADEP)

major_nonstudy_bra <- eur_bra_conns |>
  filter(!(ADES %in%  study_br)) |>   # only exclude study airports on the origin side
  group_by(ADES)  |>
  reframe(TOT_DEPS = sum(DEPS))  |>
  filter(TOT_DEPS >= conns_threshold) %>%
  pull(ADES)


eur_bra_grouped <- eur_bra_conns %>%
  mutate(
    ORIG_GROUP = case_when(
```

```r
      ADEP %in% study_eu ~ eu_labels[ADEP],
      ADEP %in% major_nonstudy_eur ~ paste0(ADEP, " (Major Non-Study)"),
      TRUE ~ "Other Europe"
    ),
    DEST_GROUP = case_when(
      ADES %in% study_br ~ br_labels[ADES],
      ADES %in% major_nonstudy_bra ~ paste0(ADES, " (Major Non-Study)"),
      TRUE ~ "Other Brazil"
    )
  )
```

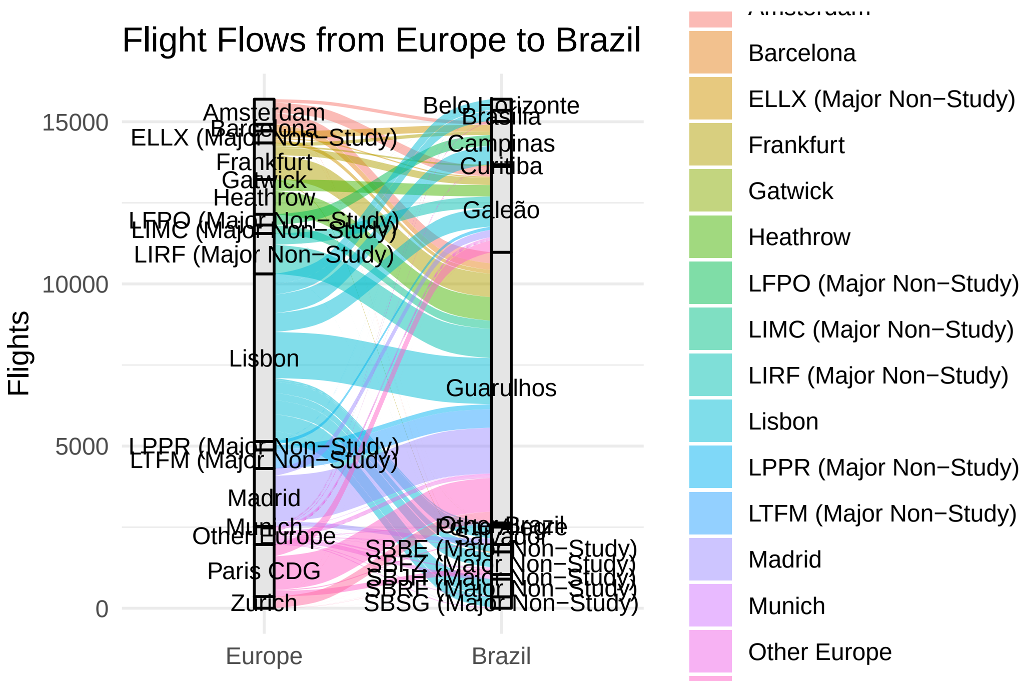
```r
library(ggalluvial)

ggplot(eur_bra_grouped, aes(axis1 = ORIG_GROUP, axis2 = DEST_GROUP, y = DEPS)) +
  geom_alluvium(aes(fill = ORIG_GROUP), width = 1/12) +
  geom_stratum(width = 1/12, fill = "grey80", color = "black") +
  geom_text(stat = "stratum", aes(label = after_stat(stratum)), size = 3.5) +
  scale_x_discrete(limits = c("Europe", "Brazil"), expand = c(.05, .05)) +
  labs(y = "Flights", title = "Main Europe-Brazil Air Connections") +
  theme_minimal()
```



```
ggplot(eur_bra_grouped,
       aes(axis1 = ORIG_GROUP, axis2 = DEST_GROUP, y = DEPS)) +
  geom_alluvium(aes(fill = ORIG_GROUP), width = 0.2, alpha = 0.8) +
  geom_stratum(width = 0.2, fill = "grey90", color = "black") +
  geom_text(stat = "stratum", aes(label = after_stat(stratum)),
            size = 3, angle = 0, hjust = 0.5, vjust = 0.5) +
  scale_x_discrete(limits = c("Origin", "Destination"), expand = c(.05, .05)) +
  theme_minimal() +
  theme(axis.text.y = element_blank(),
        axis.ticks = element_blank(),
        panel.grid = element_blank())
```




```
ggplot(eur_bra_grouped, aes(axis1 = ORIG_GROUP, axis2 = DEST_GROUP, y = DEPS)) +
  geom_alluvium(aes(fill = ORIG_GROUP), width = 1/12) +
  geom_stratum(width = 1/12, fill = "grey90", color = "black") +
  geom_text(stat = "stratum", aes(label = after_stat(stratum)), size = 3.2) +
  scale_x_discrete(limits = c("Europe", "Brazil")) +
  labs(title = "Flight Flows from Europe to Brazil", y = "Flights") +
  theme_minimal()
```



Exploration to get to alluvial plot of connections

```
library(ggforce)

library(ggplot2)
library(ggforce)
library(dplyr)

# Make sure both ORIG_GROUP and DEST_GROUP are factors (important!)
flows_grouped <- eur_bra_grouped %>%
  mutate(
    ORIG_GROUP = ifelse(ORIG_GROUP %in% study_eu, , "Study", "Non-Study"),
    DEST_GROUP = ifelse(DEST_GROUP %in% study_br,  "Study", "Non-Study")
  )

ggplot(flows_grouped) +
  geom_parallel_sets(aes(
    ORIG_GROUP,
    DEST_GROUP,
    weight = DEPS,
    fill = ORIG_GROUP
  ), alpha = 0.8) +
```

```
  # geom_parallel_sets_axes(axis.width = 0.1, fill = "gray90") +
  # geom_parallel_sets_labels(angle = 0, size = 3) +
  # scale_x_discrete(labels = c("Origin", "Destination")) +
  theme_minimal() +
  labs(title = "Flight Flows from Europe to Brazil",
       subtitle = "Grouped by Origin and Destination Airport Categories",
       y = "Number of Departures")


flows_grouped <-  eur_bra_grouped |>
  mutate(
    ORIG_STUDY = ifelse(ADEP %in% eur_apts, "Study", "Non-Study"),
    DEST_STUDY = ifelse(ADES %in% bra_apts, "Study", "Non-Study")
)
flows_long <- flows_grouped %>%
  mutate(across(c(ORIG_STUDY, DEST_STUDY), as.factor)) %>%
  gather_set_data(x = c("ORIG_GROUP", "DEST_GROUP"))

# Create the parallel sets plot with Study/Non-Study grouping
ggplot(flows_long, aes(x = x, id = id, split = y, value = DEPS)) +
  geom_parallel_sets(aes(fill = ORIG_GROUP), alpha = 0.6, axis.width = 0.1) +
  geom_parallel_sets_axes(axis.width = 0.1, fill = "gray90") +
  geom_parallel_sets_labels(angle = 0, size = 3, colour = "black") +
  scale_x_discrete(labels = c("Origin", "Destination")) +
  #scale_fill_brewer(type = "qual", palette = "Set2") +  # Use a color palette
  theme_minimal() +
  labs(
    title = "Flight Flows from Europe to Brazil (Study vs Non-Study)",
    subtitle = "Grouped by Origin and Destination Airport Categories",
    y = "Number of Departures"
  )


ggplot(flows_long, aes(x = x, id = id, split = y, value = DEPS)) +
  geom_parallel_sets(aes(fill = ORIG_STUDY), alpha = 0.6, axis.width = 0.1) +
  geom_parallel_sets_axes(axis.width = 0.1, fill = "gray90") +
  geom_parallel_sets_labels(angle = 0, size = 3, colour = "black") +
  scale_x_discrete(labels = c("Origin", "Destination")) +
  scale_fill_manual(
    values = c("Study" = "blue", "Non-Study" = "lightblue")
    ) + # ,
  #                              "Study Brazil" = "red", "Non-Study Brazil" = "lightcoral"))
  theme_minimal() +
  labs(
```

```
    title = "Flight Flows from Europe to Brazil (Study vs Non-Study)",
    subtitle = "Grouped by Origin and Destination Airport Categories",
    y = "Number of Departures"
  )
```

```
ggplot(flows_long, aes(x = x, id = id, split = y, value = DEPS)) +
  geom_parallel_sets(aes(fill = ORIG_STUDY), alpha = 0.6, axis.width = 0.1) +
  geom_parallel_sets_axes(axis.width = 0.1, fill = "gray90") +
  geom_parallel_sets_labels(angle = 0, size = 3, colour = "black") +
  scale_x_discrete(labels = c("Origin", "Destination")) +
  scale_fill_manual(values = c("Study" = "blue", "Non-Study" = "lightblue")) +
  theme_minimal() +
  labs(
    # title = "Flight Flows from Europe to Brazil (Study vs Non-Study)",
    # subtitle = "Grouped by Origin and Destination Study Status",
    y = "Number of Departures", x = element_blank(), fill = element_blank()
  ) +
  theme(legend.position = "inside", legend.position.inside = c(0.1,0.1))
```

flows_grouped <- flows_grouped %>% mutate(ORIG_LABEL = case_when(ORIG_GROUP %in% study_airports_europe ~ paste0("Study:", ORIG_GROUP), ORIG_GROUP %in% top_airports_europe ~ paste0("Non-Study:", ORIG_GROUP), TRUE ~ paste0("Other:", ORIG_GROUP)), DEST_LABEL = case_when(DEST_GROUP %in% study_airports_brazil ~ paste0("Study:", DEST_GROUP), DEST_GROUP %in% top_airports_brazil ~ paste0("Non-Study:", DEST_GROUP), TRUE ~ paste0("Other:", DEST_GROUP)))

Custom order of factor levels

all_airports <- unique(c(flows_grouped$ORIG_LABEL, flows_grouped$DEST_LABEL))

study_levels <- sort(unique(grep("^Study", all_airports, value = TRUE))) non_study_levels <- sort(unique(grep("^Non-Study", all_airports, value = TRUE))) other_levels <- sort(unique(grep("^Other", all_airports, value = TRUE)))

ordered_levels <- c(study_levels, non_study_levels, other_levels)

Apply the factor levels

flows_grouped <- flows_grouped %>% mutate(ORIG_LABEL = factor(ORIG_LABEL, levels = ordered_levels), DEST_LABEL = factor(DEST_LABEL, levels = ordered_levels))

flows_long <- flows_grouped %>% gather_set_data(x = c("ORIG_LABEL", "DEST_LABEL"))

ggplot(flows_long, aes(x = x, id = id, split = y, value = DEPS)) + geom_parallel_sets(aes(fill = ORIG_LABEL), alpha = 0.6, axis.width = 0.1) + geom_parallel_sets_axes(axis.width = 0.1, fill = "gray95") + geom_parallel_sets_labels(angle = 0, size = 3, colour = "black") + scale_x_discrete(labels = c("Origin", "Destination")) + scale_fill_manual(values = RColorBrewer::brewer.pal(9, "Set1")[1:length(study_levels)]) + theme_minimal() + labs(title = "Flight Flows: Brazil–Europe", subtitle = "Airports grouped and ordered by Study, Non-Study, Other", y = "Departures")

```
# trim connections
eur_bra_conns <- eur_mvts |>
  filter(DEP_PRU == "Eurocontrol", DES_CNTRY == "BR") |>
  group_by(ADEP, ADES) |> reframe(DEPS = n()) |> arrange(desc(DEPS))

conns_threshold <- 90

major_nonstudy_eur <- eur_bra_conns |>
  filter(!(ADEP %in%  study_eu)) |>   # only exclude study airports on the origin side
  group_by(ADEP)  |>
  reframe(TOT_DEPS = sum(DEPS))  |>
  filter(TOT_DEPS >= conns_threshold) %>%
  pull(ADEP)

major_nonstudy_bra <- eur_bra_conns |>
  filter(!(ADES %in%  study_br)) |>   # only exclude study airports on the origin side
  group_by(ADES)  |>
  reframe(TOT_DEPS = sum(DEPS))  |>
  filter(TOT_DEPS >= conns_threshold) %>%
  pull(ADES)

eur_bra_grouped <- eur_bra_conns %>%
  mutate(
    ORIG_NAME = case_when(
      ADEP %in% study_eu ~ eu_labels[ADEP],
      ADEP %in% major_nonstudy_eur ~ ADEP,
```

```
      TRUE ~ "Other Europe"
    ),
    DEST_NAME = case_when(
      ADES %in% study_br ~ br_labels[ADES],
      ADES %in% major_nonstudy_bra ~ ADES,
      TRUE ~ "Other Brazil"
    )
  )

# Make sure both ORIG_GROUP and DEST_GROUP are factors (important!)
flows_grouped <- eur_bra_grouped %>%
  mutate(
    ORIG_GROUP = ifelse(ADEP %in% eur_apts, "Study", "Non-Study"),
    DEST_GROUP = ifelse(ADES %in% bra_apts,  "Study", "Non-Study")
  )

# prep for parallel sets plot
flows_long <- flows_grouped |>
  # need to gather to feed set_parallel !!!!
  gather_set_data(x = c("ORIG_NAME","DEST_NAME")) |>
#add other stuff
mutate(GROUP_TYPE = case_when(
    ORIG_GROUP == "Study" ~ "blue"
  , DEST_GROUP == "Study" ~ "blue"
  , .default = "gray80"))

## Step 1: Get unique y/group_type mapping
# axis_fill_map <- flows_long %>%
#   select(y, GROUP_TYPE) %>%
#   distinct()

# Step 2: Join back to deduplicated mapping
# flows_long_fixed <- flows_long %>%
#   select(-GROUP_TYPE) %>%
#   left_join(axis_fill_map, by = "y")
```

write out prep results

flows_long |> mutate(YEAR = 2024) |> write_csv(“./data/BRA-EUR-conns-parallel.csv”)

```
flows_long <- read_csv(here::here("data","BRA-EUR-conns-parallel.csv"), show_col_types = FALS
```

```r
ggplot(flows_long, aes(x = x, id = id, split = y, value = DEPS)) +
  geom_parallel_sets(aes(fill = ORIG_GROUP), alpha = 0.6, axis.width = 0.1) +
  geom_parallel_sets_axes(axis.width = 0.2, fill = "gray90") +
  geom_parallel_sets_labels(angle = 0, size = 3, colour = "black") +
  scale_x_discrete(labels = c("Origin", "Destination")) +
  scale_fill_manual(values = c("Study" = "blue", "Non-Study" = "lightblue")) +
  theme_minimal() +
  labs(
    # title = "Flight Flows from Europe to Brazil (Study vs Non-Study)",
    # subtitle = "Grouped by Origin and Destination Study Status",
    y = "Number of Departures", x = element_blank(), fill = element_blank()
  ) +
  theme(legend.position = "inside", legend.position.inside = c(0.1,0.1))
```

```
Warning: `label` cannot be a <ggplot2::element_blank> object.
`label` cannot be a <ggplot2::element_blank> object.
```